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Association between Perceived Stress and Metabolic Parameters: A Systematic Review and Meta-Analysis

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ABSTRACT

Background

Metabolic syndrome (MetS) is a major global public health problem associated with cardiovascular disease, type 2 diabetes mellitus (T2D), and increased mortality. Chronic psychological stress has been proposed as a contributor to metabolic dysfunction; however, evidence linking perceived stress to MetS and its individual components remains inconsistent. The Perceived Stress Scale (PSS) is a widely validated instrument for assessing subjective stress, yet its association with metabolic outcomes has not been systematically synthesised.

Aims

The objective of this study was to systematically review the evidence and evaluate the association between perceived stress, measured using the PSS, and metabolic parameters related to MetS.

Methods

A systematic search of peer-reviewed literature identified studies assessing perceived stress using the PSS-10 or PSS-14 and reporting metabolic outcomes in adults. Associations between perceived stress and metabolic parameters were examined using random-effects meta-regression models, with stress scores standardised to allow comparability across PSS versions. Outcomes included anthropometric measures, blood pressure, glycaemic markers, lipid profile, and insulin resistance.

Results

Thirty studies comprising 12,491 participants were included. Most metabolic parameters, including body mass index, waist circumference, blood pressure, fasting glucose, and triglycerides, showed no significant association with perceived stress. In contrast, insulin resistance assessed by HOMA-IR was positively associated with perceived stress ($\beta = 0.463$, 95% CI: 0.159–0.767). Borderline associations were observed for high-density lipoprotein cholesterol and total cholesterol. Although substantial heterogeneity was present across analyses, the HOMA-IR association was exploratory and sensitive to the inclusion of individual studies.

Conclusions

Higher perceived stress may be associated with insulin resistance (HOMA-IR) but not with traditional anthropometric, glycaemic, or blood pressure measures; this finding is exploratory and causal inferences cannot be established. These results highlight the potential relevance of psychosocial stress as a contributor to metabolic vulnerability; stress assessment may be considered in preventive and clinical strategies pending higher-certainty evidence.

INTRODUCTION

Metabolic syndrome (MetS) is a multifactorial clinical condition characterised by central obesity, dyslipidaemia, hypertension, insulin resistance, oxidative stress, and chronic low-grade inflammation.[1] It represents a major global public health challenge due to its strong association with cardiovascular disease, type 2 diabetes mellitus (T2D), and increased all-cause mortality.[2] Over recent decades, its prevalence has increased substantially, reflecting widespread changes in lifestyle behaviours and psychosocial environments. Although lifestyle modification remains the cornerstone of prevention and management, the complexity of MetS aetiology highlights the need to identify additional modifiable risk factors and complementary prevention strategies.[3]

Chronic psychological stress has been proposed as a key contributor to metabolic dysfunction.[4] Biological evidence suggests that the appraisal and experience of stress triggers a cascade of physiological stress responses; when psychological stress is sustained,[5] it leads to prolonged activation of stress-related neuroendocrine pathways, particularly the hypothalamic–pituitary–adrenal (HPA) axis and the sympathetic nervous system, that may promote visceral adiposity, insulin resistance, systemic inflammation, and adverse lipid profiles. These mechanisms provide a plausible physiological link between sustained psychological stress and the development of MetS.[6]

Despite its relevance, chronic stress remains challenging to quantify in epidemiological research. Self-reported questionnaires remain the most common method, and among them, the Perceived Stress Scale (PSS) is one of the most widely used and validated instruments for assessing subjective stress perception.[7] Unlike objective stressor counts, the PSS captures an individual's appraisal of stress, which may more closely reflect stress-related neuroendocrine activation and long-term health outcomes.[8]

Several observational studies have explored the association between perceived stress measured by the PSS and MetS or its individual components; however, findings have been inconsistent.[9,10] Heterogeneity in study design, population characteristics, PSS versions and cutoffs, and MetS diagnostic criteria has limited comparability across studies.[9,10] Previous reviews have often combined diverse stress measures, further weakening the interpretability of pooled estimates.[3]

To date, no systematic review and meta-analysis has specifically examined the association between chronic perceived stress measured exclusively with the PSS and metabolic syndrome-related outcomes. Therefore, this study aimed to systematically review the available evidence and conduct a meta-regression analysis to evaluate the relationship between perceived stress and metabolic parameters.

METHODS

Search Strategy and Study Selection

This systematic review and meta-analysis were conducted in accordance with the PRISMA 2020 guidelines (see S1 Checklist) and a predefined protocol registered in PROSPERO (CRD420251231160).

A systematic literature search was conducted in PubMed and EMBASE to identify studies examining the association between perceived stress and metabolic risk factors. The same search strategy was adapted for both databases; after deduplication against the PubMed results, two records not retrieved by PubMed underwent full-text review, of which one (Tan and Leung)[28] met all eligibility criteria and was added, bringing the total to 30 included studies. Only studies published from 2015 onward were considered; this date threshold was determined to focus on evidence generated after widespread adoption of the PSS-10 as the dominant instrument in metabolic research, and is acknowledged as a limitation.

The PubMed search query was: (“Perceived Stress Scale”[Title/Abstract] OR “perceived stress”[Title/Abstract] OR PSS[Title/Abstract]) AND (“metabolic syndrome”[Title/Abstract] OR “insulin resistance”[Title/Abstract] OR “HOMA-IR”[Title/Abstract] OR “obesity”[Title/Abstract] OR “body mass index”[Title/Abstract] OR “waist circumference”[Title/Abstract] OR “hypertension”[Title/Abstract] OR “dyslipidaemia”[Title/Abstract]). No restrictions were placed on geographic location or participant characteristics. The literature search was conducted from October 2025 to January 2026.

Data Extraction

Data extraction was performed independently by two reviewers. From each included study we extracted: first author name, publication year, sample size, PSS version (10-item or 14-item), PSS mean and standard deviation, and means and standard deviations for all available metabolic parameters including BMI, waist circumference, SBP, DBP, fasting glucose, HOMA-IR, HbA1c, total cholesterol, HDL, LDL, triglycerides, and CRP.

Data Preparation and Standardisation

Although the PROSPERO protocol specified inclusion of studies using the PSS-10, studies employing the PSS-14 were also included to maximise available evidence. To ensure comparability across instruments, perceived stress scores were standardised using z-scores: z = (study PSS mean − overall PSS mean) / overall PSS SD, where the overall mean was 19.20 (SD = 9.48) across all 30 studies. Variance of the z-score was calculated using propagation of error: $Var(z) = (\text{study PSS SD})^2 / (\text{sample size} \times \text{overall PSS SD}^2)$.

Eligibility Criteria

Studies were eligible if they: (1) used an observational or interventional design reporting baseline associations; (2) enrolled adults aged ≥18 years; (3) assessed chronic psychological stress with PSS-10 or PSS-14; (4) reported MetS diagnosis or at least one metabolic component; and (5) provided sufficient data to derive effect estimates. Case reports, narrative reviews, editorials, conference abstracts, paediatric studies, and studies using non-PSS instruments were excluded.

Risk of Bias Assessment

Randomised controlled trials were assessed using the Cochrane Risk of Bias 2 (RoB 2) tool. Observational studies were assessed using the Newcastle-Ottawa Scale (NOS), categorised as low (7–9), moderate (5–6), or high (≤4) risk of bias. Full assessments are presented in the supplemental material (NOS) and S4 (RoB 2).

Statistical Analysis

Univariate meta-regression analyses examined the association between each metabolic risk factor and PSS z-scores using the rma.mv function (REML) in the metafor package (v4.6-0) in R (v4.4.2). Between-study heterogeneity was quantified with I^2, τ^2, and Cochran's Q-test. Publication bias was assessed by Egger's regression test (for analyses with $k \geq 10$; Egger's test was not interpreted for outcomes with $k < 10$, for which Orwin's Fail-Safe N is reported as a supplementary robustness check). Sensitivity analyses included leave-one-out re-estimation for all primary outcomes, moderator analyses by study design and population type, and re-running all models in the protocol-aligned direction (PSS z-score as predictor of each metabolic outcome), reported in supplemental material. Statistical significance was set at $\alpha = 0.05$; p-values 0.05–0.10 were considered borderline significant.

Ethics Statement

This study is based on previously published data and does not involve human participants; therefore, ethical approval was not required.

RESULTS

Study Selection

The study selection process is summarised in Figure 1.

Study Characteristics and Data Availability

A total of 30 studies met the inclusion criteria, comprising 12,491 participants. Of these, 16 (53.3%) used the PSS-10 and 14 (46.7%) the PSS-14. The mean PSS score across all studies was 19.20 (SD = 9.48). Figure 2 displays the distribution of PSS scores by version. BMI was the most frequently reported variable (24 studies), followed by systolic and diastolic blood pressure (12 studies each), and waist circumference (10 studies). Other metabolic markers were less commonly reported: glucose in 7 studies, triglycerides and HbA1c in 5 each, HOMA-IR and HDL in 4 each, CRP and total cholesterol in 3 each, and LDL in only 2 studies. Standardised PSS z-scores stratified by PSS version and sample size are shown in Figure 3.

Univariate Meta-Regression Analyses

Table 2 presents the results of univariate meta-regression analyses examining metabolic risk factors as predictors of standardised PSS scores across the combined dataset. Most metabolic variables showed no significant association with perceived stress. BMI ($\beta = 0.0084$, 95% CI: −0.079 to 0.095, $p = 0.844$), waist circumference ($\beta = 0.019$, 95% CI: −0.037 to 0.076, $p = 0.453$), fasting glucose ($\beta = 0.019$, 95% CI: −0.078 to 0.115, $p = 0.641$), triglycerides ($\beta = -0.027$, 95% CI: −0.075 to 0.020, $p = 0.165$), systolic blood pressure ($\beta = -0.036$, 95% CI: −0.073 to 0.001, $p = 0.057$), and diastolic blood pressure ($\beta = -0.019$, 95% CI: −0.105 to 0.066, $p = 0.624$) were not significantly associated with PSS z-scores. Characteristics of included studies are summarised in Table 1.

However, HOMA-IR showed a significant positive association with perceived stress ($\beta = 0.463$, 95% CI: 0.159 to 0.767, $p = 0.022$, $k = 4$ studies; $I^2 = 95.1\%$, $\tau^2 = 0.053$; Figures 4, 4b). This association should be interpreted as exploratory given the small number of contributing studies and the extreme heterogeneity; leave-one-out analysis (supplemental material) showed it was sensitive to the removal of individual studies.

This indicates that a one standard deviation increase in the PSS z-score was associated with a 0.463-unit increase in HOMA-IR, suggesting that higher perceived stress levels are linked to greater insulin resistance.

Two additional variables showed borderline significant associations: HDL cholesterol demonstrated a borderline negative association ($\beta = -0.080$, 95% CI: −0.166 to 0.006, $p = 0.057$, $k = 4$ studies), and total cholesterol showed a borderline positive association ($\beta = 0.037$, 95% CI: −0.012 to 0.086, $p = 0.066$, $k = 3$ studies; Figure 4), though these results should be interpreted cautiously given the marginal p-values and small number of contributing studies.

Heterogeneity

Very high heterogeneity was observed across most meta-regression analyses (Table 2). The I^2 statistic exceeded 90% for all metabolic predictors except total cholesterol ($I^2 = 57.7\%$), indicating that the vast majority of variance in observed associations was due to between-study differences rather than sampling error. For the significant HOMA-IR association, heterogeneity was high ($I^2 = 95.1\%$, $\tau^2 = 0.053$). The Q-test was significant ($p < 0.001$) for all analyses.

Risk of Bias

Risk of bias assessments are summarised in Supplementary Tables S3 (NOS, observational studies) and S4 (RoB 2, randomised controlled trials). Among observational studies, most were judged to be at low risk of bias. Among RCTs, most were judged as having some concerns, primarily due to the lack of blinding inherent to behavioural interventions, incomplete outcome data, and the risk of selective outcome reporting. A smaller subset of RCTs was judged to be at high risk of bias due to substantial attrition and/or deviations from intended interventions.

Publication Bias and Robustness Analyses

Because formal funnel-plot-based tests are not reliable when fewer than 10 studies contribute, Egger's test could not be meaningfully applied to HOMA-IR ($k = 4$) or to any other outcome with $k < 10$ (including HDL, total cholesterol, and CRP). For HOMA-IR, Orwin's Fail-Safe N indicated that 32 unpublished null-result studies would be required to reduce the observed association to a trivial level ($r < 0.05$; missing-to-observed ratio 8.12:1); this is presented as a supplementary robustness check rather than a formal assessment of publication bias. With a more stringent trivial threshold of 0.10, 14 null studies would still be required. Egger's regression test for BMI ($k = 24$) was not statistically significant ($t = 1.13$, $df = 8$, $p = 0.289$), suggesting no strong evidence of funnel plot asymmetry. The correlation between sample size and PSS z-scores was weak and non-significant ($r = -0.074$, $p = 0.738$; Table 3).

Sensitivity and Multivariate Analyses

Leave-one-out sensitivity analyses for the BMI meta-regression demonstrated stable results (beta coefficients ranging from 0.010 to 0.021 across iterations). Similar sensitivity analyses were conducted for other primary metabolic predictors, all showing consistent results. All meta-regression models were also re-run in the protocol-aligned direction (PSS z-score predicting each metabolic outcome); results were consistent and are reported in the supplemental material. For HOMA-IR, the leave-one-out analysis (supplemental material) showed that the association remained significant when Raja-Khan (2017) was omitted ($k = 3$: $\beta = 2.072$, 95% CI: 0.189–3.955, $p = 0.045$) but became non-significant when any of the other three studies (Berg-Schmidt, Tauqir, or Kautzky) was removed, confirming that this finding is exploratory and requires replication.

Certainty of Evidence

According to the GRADE framework, the certainty of evidence for all primary outcomes was rated as very low (Table 4). Although a statistically significant association was observed between HOMA-IR and perceived stress, the certainty was downgraded due to the observational study design, high heterogeneity ($I^2 > 90\%$), and the limited number of contributing studies.

DISCUSSION

Metabolic syndrome is a complex disorder characterised by a cluster of clinical and biochemical abnormalities that contribute to the early development of metabolic and cardiovascular diseases.[1] Its increasing global prevalence has expanded research beyond traditional risk factors, highlighting the role of non-classical determinants such as chronic psychological stress. These observations are consistent with the broader evidence base linking psychological stress to obesity and its cardiometabolic complications.[101129]

Chronic stress represents a disruption of homeostasis induced by psychological, physiological, or environmental stimuli. While acute stress responses are adaptive, sustained activation of stress pathways leads to neurochemical, neuroanatomical, and cellular changes that adversely affect multiple physiological systems.[12] Central to this response is activation of the HPA axis and the sympathetic nervous system, resulting in increased cortisol secretion and catecholamine release.[13]

Glucocorticoids exert counter-regulatory effects on insulin, promoting gluconeogenesis, lipolysis, and energy storage.[14] Cortisol preferentially stimulates visceral adipose tissue, which exhibits higher glucocorticoid receptor density than subcutaneous fat, contributing to central fat accumulation, increased free fatty acid release, and inflammatory signalling, all of which promote insulin resistance.[15]

These mechanisms are further supported by clinical and experimental models in which excess glucocorticoid exposure induces insulin resistance, such as in Cushing syndrome or exogenous glucocorticoid administration,[16,17] reinforcing the biological plausibility of a link between chronic stress and impaired insulin sensitivity.

In this systematic review and meta-regression, insulin resistance emerged as the primary metabolic correlate of perceived stress, whereas traditional anthropometric and glycaemic measures were not significantly associated. This finding aligns with biological evidence linking chronic stress, HPA axis hyperactivity, and functional hypercortisolism to impaired insulin sensitivity.[18,19] Notably, the lack of significant associations with BMI, waist circumference, and fasting glucose suggests that stress-related metabolic risk may manifest initially through functional metabolic alterations rather than overt obesity or hyperglycaemia.[20] HOMA-IR, which captures both insulin sensitivity and compensatory hyperinsulinaemia, may therefore represent a more sensitive marker of early stress-related metabolic dysregulation.[21] This is further supported by evidence indicating that stress may exacerbate glycaemic responses particularly in individuals with pre-existing insulin resistance.[22] Collectively, these findings suggest that perceived stress may contribute to early metabolic vulnerability, particularly at the level of insulin signalling.[21,22] These changes may precede the development of clinically overt metabolic syndrome.[20]

However, evidence directly linking perceived stress to neuroendocrine alterations remains heterogeneous. For example, no significant changes in diurnal salivary cortisol patterns have been observed following stress-reduction interventions despite changes in perceived stress levels, whereas higher long-term cortisol exposure (measured via hair cortisol) has been reported in individuals with type 2 diabetes.[23,24] These findings suggest that the relationship between perceived stress and physiological stress markers is complex and may not be fully captured by single time-point assessments.

From a clinical perspective, chronic stress may act as a modifier of metabolic risk, particularly in individuals with underlying susceptibility. Emerging evidence suggests that stress may worsen glycaemic control and metabolic outcomes in individuals with pre-existing insulin resistance, supporting its role as a relevant and potentially modifiable risk factor.[22]

In addition to neuroendocrine mechanisms, chronic stress is associated with behavioural and lifestyle factors, such as physical inactivity, poor dietary patterns, sleep disturbances, smoking, and alcohol use, that may further exacerbate metabolic dysregulation.[25] Longitudinal evidence supports this relationship, showing that individuals exposed to higher cumulative life stress exhibit progressive deterioration in lipid profiles and insulin sensitivity over time.[4]

Interventional studies further support the potential reversibility of stress-related metabolic alterations. Mindfulness-based stress reduction interventions have demonstrated improvements in fasting glucose levels,[26] while stress-reduction strategies such as yoga have shown beneficial effects on blood pressure and perceived stress levels.[27] These findings reinforce the potential clinical relevance of addressing stress as part of a comprehensive metabolic risk management strategy.

Limitations

This study has several limitations. High heterogeneity across analyses reflects substantial variability in study design, populations, stress assessment, and covariate adjustment. I^2 exceeded 90% for most outcomes, and exploratory moderator analyses (by PSS version, study design, and clinical vs. general population) did not explain this heterogeneity appreciably; pooled estimates should therefore be regarded as hypothesis-generating. The predominance of cross-sectional studies limits causal inference, and the small number of studies available for certain metabolic predictors reduces statistical power. Residual confounding and publication bias cannot be entirely excluded.

Although inclusion of both PSS-10 and PSS-14 enhanced study coverage, this may have contributed to between-study heterogeneity despite score standardisation. Consistent with these limitations, the overall certainty of evidence was rated as very low according to the GRADE framework.

Implications and Future Directions

These findings suggest that insulin resistance may represent an early metabolic signature linked with chronic psychological stress, potentially preceding the development of overt metabolic disease. From a clinical and public health perspective, this highlights the importance of early identification and monitoring of individuals experiencing high levels of perceived stress, even in the absence of established metabolic syndrome.

Future high-quality longitudinal and interventional studies are warranted to clarify temporal relationships and to determine whether reducing perceived stress can improve insulin sensitivity and ultimately reduce the risk of metabolic disease.

CONCLUSIONS

This systematic review and meta-analysis provide very low certainty evidence that higher levels of perceived stress are associated with insulin resistance (HOMA-IR), but not with traditional anthropometric, glycaemic, or blood pressure measures; because this finding rests on four studies with extreme heterogeneity and is sensitive to the removal of individual studies, it should be interpreted as hypothesis-generating, and causal inferences cannot be established.

However, the overall certainty of evidence is very low due to the observational nature of the included studies, substantial between-study heterogeneity, and the limited number of studies available for certain outcomes.

Despite these limitations, the observed association highlights the potential relevance of psychosocial stress as a contributor to early metabolic dysregulation. Incorporating stress assessment into clinical practice may support preventive strategies and early risk stratification. Further high-quality longitudinal and interventional studies are needed to clarify temporal relationships and determine whether reducing perceived stress can improve insulin sensitivity and ultimately reduce the risk of metabolic disease.

DISCLOSURES

Data availability

All relevant data and code are available within the manuscript and its Supporting Information files.

Competing interests

The authors have declared that no competing interests exist.

Funding

This study did not receive external funding.

Supporting Information

S1 Checklist. PRISMA 2020 checklist.
S2 File. R code used for data cleaning, standardisation of PSS scores, and meta-regression analyses.
S3 Table. Risk of bias assessment for observational studies (Newcastle-Ottawa Scale).
S4 Table. Risk of bias assessment for randomised controlled trials (Cochrane RoB 2).
S5 Table. Protocol-aligned meta-regression results (PSS z-score predicting each metabolic outcome).
S6 Table. Leave-one-out sensitivity analysis for the HOMA-IR meta-regression.

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FIGURE LEGENDS

Fig 1. PRISMA flow diagram of study selection.
Fig 2. Distribution of PSS scores by scale version.
Fig 3. Standardised PSS z-scores across studies.
Fig 4. Meta-regression coefficients showing how stress predicts metabolic dysfunction. Note: this figure displays results from the protocol-aligned sensitivity analysis (PSS z-score as predictor of each metabolic outcome; supplemental material). Triglycerides appears in red ($p < 0.05$) in this direction; in the primary analysis (metabolic variable predicting PSS z-score, Table 2), Triglycerides is not significant ($p = 0.165$).

Fig 4b. Forest plot of the four studies contributing to the HOMA-IR meta-regression, showing individual study means and 95% confidence intervals with the random-effects pooled estimate ($I^2 = 99.4\%$; note: this I^2 reflects heterogeneity in raw study-level HOMA-IR means from a simple random-effects meta-analysis and differs from the meta-regression heterogeneity statistic $I^2 = 95.1\%$ reported in Table 2).

TABLES

Table 1. Characteristics of included studies and reported metabolic variables.

First Author	Year	N	PSS	PSS Mean	PSS SD	BMI	Waist (cm)	Glucose
Atanes	2015	450	PSS-14	42.20	13.90	—	—	—
Palagini	2015	330	PSS-10	14.63	8.65	—	—	—
Järvelä-Reijonen	2016	297	PSS-14	26.50	7.70	31.30	102.86	—
Choudhary	2017	50	PSS-14	20.13	4.10	27.02	—	—
Fontvieille	2017	31	PSS-10	13.36	4.66	29.10	—	—
Raja-Khan	2017	86	PSS-10	22.19	3.00	38.88	111.30	105.00
Berg-Schmidt	2018	47	PSS-10	15.15	5.77	22.30	72.60	92.43
Morin	2018	60	PSS-14	14.92	5.22	32.00	99.03	—
Payne	2018	67	PSS-10	10.50	5.90	36.90	—	—
Pearl	2018	137	PSS-14	20.60	8.50	40.80	—	—
Soltani	2018	44	PSS-10	12.70	2.29	—	—	—
Stinson	2018	46	PSS-14	4.20	2.20	28.30	—	92.70
Watson	2018	61	PSS-10	14.01	7.00	34.30	—	—
Chang	2019	569	PSS-10	2.56	0.55	32.04	—	—
D'Alonzo	2019	59	PSS-14	32.50	5.59	30.10	—	—

Wilson	2020	53	PSS-14	21.80	8.80	—	—	—
Kautzky	2021	43	PSS-10	15.59	5.32	27.60	97.99	—
Tan	2021	8385	PSS-14	22.70	6.20	24.30	—	—
Tauqir	2021	147	PSS-14	34.20	4.80	29.66	102.05	97.64
Jeitler	2022	145	PSS-14	13.70	3.30	33.30	113.10	111.70
Koch	2022	120	PSS-10	29.77	3.60	—	—	—
Hassan	2023	110	PSS-14	34.00	5.00	33.48	105.42	—
Pathan	2023	64	PSS-10	20.27	4.60	25.62	—	—
Wong	2023	264	PSS-10	14.91	5.07	26.90	93.85	108.10
Bellach	2024	41	PSS-14	15.18	4.77	28.00	—	—
Blumenthal	2024	140	PSS-10	10.49	7.18	36.00	—	—
Akçin	2025	100	PSS-10	27.10	7.10	—	—	—
Jin	2025	74	PSS-10	2.86	1.41	24.70	—	—
Santos	2025	91	PSS-14	22.78	9.41	28.32	—	124.00
Tawfik	2025	380	PSS-10	24.50	9.51	26.50	91.95	—

— indicates data not available for that study

Table 2. Meta-regression results: effect estimates, heterogeneity, and significance for metabolic predictors of perceived stress.

PSS Version	**Metabolic Predictor**	**k**	**β**	**95% CI**	**p-value**	**Sig.**	**I²**	**Heterogeneity**
PSS-10	BMI	12	−0.0050	[−0.0984, 0.0884]	0.9077		99.9%	High
PSS-10	Waist Circumference	5	0.0153	[−0.0415, 0.0720]	0.4551		99.1%	High
PSS-10	Glucose	3	0.0157	[−0.6247, 0.6562]	0.8073		99.6%	High

PSS-10	HDL	3	−0.0817	[−0.5854, 0.4220]	0.2875		98.1%	High
PSS-10	Triglycerides	3	−0.0042	[−0.3539, 0.3455]	0.9028		99.7%	High
PSS-14	BMI	12	0.0183	[−0.1429, 0.1796]	0.8053		99.8%	High
PSS-14	Waist Circumference	5	−0.0548	[−0.4028, 0.2931]	0.6504		99.8%	High
PSS-14	Glucose	4	0.0189	[−0.2656, 0.3035]	0.8016		99.9%	High
Combined	BMI	24	0.0084	[−0.079, 0.095]	0.8439		99.9%	High
Combined	Waist Circumference	10	0.0197	[−0.038, 0.077]	0.4533		99.7%	High
Combined	Glucose	7	0.0189	[−0.079, 0.117]	0.6414		99.9%	High
Combined	Total Cholesterol	3	0.0378	[−0.012, 0.087]	0.0656	†	57.7%	Moderate
Combined	HDL	4	−0.0810	[−0.168, 0.006]	0.0570	†	97.4%	High
Combined	Triglycerides	5	−0.0276	[−0.076, 0.020]	0.1651		99.6%	High
Combined	Systolic BP	12	−0.0362	[−0.074, 0.001]	0.0570	†	99.5%	High
Combined	Diastolic BP	12	−0.0196	[−0.106, 0.067]	0.6237		99.6%	High

Combined	HOMA-IR	4	0.4629	[0.159, 0.767]	0.0225	*	95.1%	High
Combined	HbA1c	5	−0.5594	[−1.313, 0.195]	0.0993	†	99.4%	High
Combined	CRP	3	0.6596	[−0.650, 1.969]	0.0987	†	90.3%	High

* $p<0.05$, † $p<0.10$. I^2: Low <50%, Moderate 50-75%, High ≥75%. Total Studies: 30; Total N: 12,491.

Table 3. Publication bias assessment.

Predictor	**k**	**Correlation (r)**	**p-value**	**Small Study Bias Detected**
BMI	24	−0.074	0.738	No
Significant correlation (p<0.05) suggests potential small study bias.				

Table 4. Certainty of evidence (GRADE) for the association between perceived stress and metabolic outcomes.

Outcome	**k**	**Effect estimate**	**Risk of bias**	**Inconsistency**	**Imprecision**	**Certainty**
HOMA-IR	4	β=0.463 (p=0.022)	Serious	Very serious (I^2=95.1%)	Serious	Very low
BMI	24	Not significant	Serious	Very serious	Serious	Very low

Waist circumference	10	Not significant	Serious	Very serious	Serious	Very low
Fasting glucose	7	Not significant	Serious	Very serious	Serious	Very low
Triglycerides	5	Not significant	Serious	Very serious	Serious	Very low

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